CHAPTER 1

CHAPTER 1

INTRODUCTION

The portable embedded system increases vehicle safety by continuously monitoring essential parameters of wheel alignment-camber, caster, and toe angles-along with tire pressure using a compact sensor-based setup in real-time. An ESP32-C3 Sender Module communicates with the sensors, which include an ADXL345 accelerometer, an AS5600 magnetic rotary sensor, and an MPX5700AP pressure sensor, transmits the acquired measurements wirelessly to an ESP32 receiver through low-latency wireless protocol. Then, the receiver compares these measured readings with predefined thresholds; for instance, camber will be about ±1°, toe will be about ±0.5°, and the tire pressure is within the normal operating band, displays the alignment and pressure status on OLED, generating appropriate alerts toward misalignment or unsafe conditions of tires that would help reduce accidents, uneven wear of tires, and fuel inefficiency. The Li-ion battery-powered breadboard prototype, supported by a dedicated charging and boost converter module, thus becomes a low-cost portable solution to expensive conventional bulky machines used in workshops with the capability of getting further improved by connectivity to mobile applications, cloud logging of data, and AI-based predictive maintenance.

1.1 Preamble:

A number of mechanical conditions determine the stable operation of the vehicles, but the most serious and in the vast majority of cases, the most neglected is the wheel alignment and the health of tires. A small variation in the angle of the wheels or reduction in the pressure may compromise the distance braking, steering and the fuel consumption and the safety. of the tires. This loophole led to the need of the system that would be capable of continuously monitoring the wheel orientation and the tire condition within the car without necessarily accessing a working room. It is through this that our project is meant to come up with a Smart Wheel Misalignment and Tire-Health Monitoring System, which has the capability of resisting real time information in a small, compact and affordable package that can be used by the ordinary driver as well as the automotive technician.

This project has a conceptual concept i.e. to remove these early fault-detections out of the workshops and to place them in the hands of the users. To achieve this, sensors that have control modules are included in the design that monitor the tilt of the individual wheels, steering behavior and tire pressure change. The processing of these parameters is done by a microcontroller platform and the information is displayed or transferred wirelessly to be interpreted on the spot.

One of the strengths of this project is a flexible and practical hardware architecture. The layout in the department report which was previously used formed the starting position, only some adjustments were made when implemented on our side to make it more reliable, power efficient and practical. We are improving the design with ESP32-C3 that is miniature, consumes fewer energy and it has an inbuilt connection option.

The batteries are used as the storage component of the system which works in portable mode and uses rechargeable Li-ion batteries. To achieve in fluctuated loading, the project will be regulated by a combination of reliable elements of power, TP4056 charging modules, and MT3608 boost converters that will also be employed to ensure that the project achieves in terms of safety of charging, regulated output and adjustable levels of voltages to necessities of the sensors and regulators.

In order to address this, the project entailed the decoupling of capacitors between the supply lines to suppress (voltage) variation and guard the controller and sensors against transients. They have resistors, switching, jumper wires, and safe signal conditioning as well as calibration and simplicity of testing. They are all serviceable and modular modules enclosure that can be replaced easily without having to interfere with the rest of the electronics, which is beneficial to systems where hard driving is to be used. The automotive technology can appear at times to be too complicated to be employed by a common user, thus, the idea was to eliminate any unwarranted complication. It has been noted that another important feature of vehicle functioning is how installed systems provide an oversight of functioning.

Every single technological unit has a designated role it has to play to help the unit of the vehicle work without issues. The sensors and fitting units are playing their monitoring roles earnestly to provide what the operation requires and the core processing units help them by exhibiting speed and doing processes and learning in their digital institutions. The fitted devices are often easily position able and need not be subject to permanent adjustments that make it easy to the user. Within a properly operating system the consideration of all elements in the system are accorded value and decisions are then made through a trade-off between technical complexity and the ease of usage.

The development unit also teaches us the topic of demonstrating procedural ways and possessing understanding of underlying difficulties. It is considered that there is a perfect system since many people believe that there is no place that disagreements and confusions cannot occur any time. In a project where the design is done with a lot of care, developers are trained on how to examine the reality in the problem of misalignment and tyre failure and proceed with the development.

The system provides a user with a stable ground on which to have feelings of accuracy and also the motivation to proceed. Analysts assert that the belief that the system will be successful is in some instances the greatest energy that drives such an individual to turn into real time detection, movability and straightforward data analysis into a fact.

One can say that it is the diagnostic system which is the very heart of the vehicle existence. It is here in the system where we find the analysis and direction and safety and calm. The completed prototype shows the functionality that the system is supposed to possess and complies with critical metric observation. Our knowledge consists of our monitoring units and teaches us important teachings about technical issues and provides us with constant assistance concerning maintenance.

A happy system develops a happy maintenance program and a happy maintenance program creates a better world to all. Due to this we must at all times cherish and respect and take care of our observations possibilities as they are the actual precious things that we have in our car life.

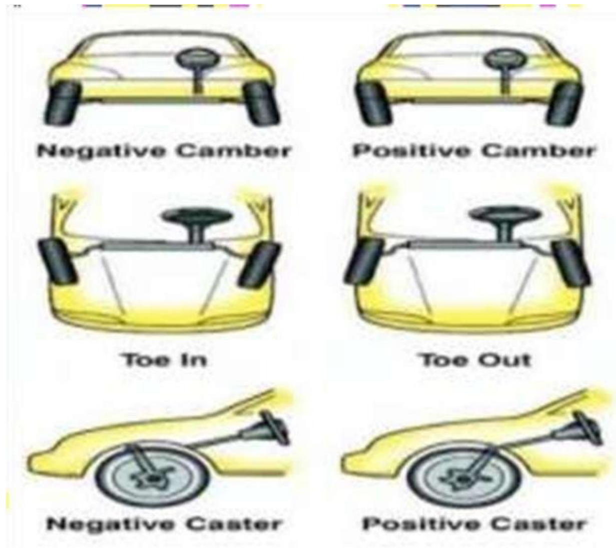


Figure 1.1 Misalignment angle Conditions

1.2 Motivation:

When it comes to the safety of a vehicle and general comfort when driving, wheel alignment and the state of tires are of significant importance. The smallest misalignment can lead to inequal wear of the tires, imprecise handling, increased fuel usage and malfunctioning. The car becomes unstable when the wheels are always misaligned over extended durations, which is likely to cause accidents particularly during sudden braking or turns. On the same note, inaccurate tier pressure influences grip, braking distance and ways that tyres last, rendering the vehicle unsafe to drive.

Although these are some of the risks, majority of drivers will always resort to checking with alignment only when the major developed symptoms occur, simply because the conventional alignment equipment is costly and bulky. Frequent checks on alignment are not feasible among most vehicles or small garages and minor problems normally go unattended until they cause larger problems.

This is the reason as to why I have been motivated towards a basic, low-cost system that is able to verify wheel alignment and tyre pressure within a fixed stationary system within a short period of time. The project is meant to assist users in detecting misalignment at the initial stages, saving on expenditure on maintenance, and enhancing further automobile security without depending on bulky workshop equipment.

1.3 Problem Statement:

With the increasing incidence of uneven tire wear, reduced fuel efficiency, and compromised vehicle safety due to undetected wheel misalignment and poor tire health, there is a great need for an intelligent wheel misalignment and tire health monitoring system in automobiles. The current manual inspection methods are infrequent, tedious, and highly dependent on the expertise of the technician, and therefore often result in the late identification of accident - causing conditions, leading to high maintenance expenses. The idea of this project is to develop an intelligent, real-time monitoring solution capable of continuous measurement of wheel alignment parameters and key tire health indicators in respect of wear, pressure, and temperature, while warning the driver against abnormal conditions. By providing early fault detection and actionable information, the system will aim to enhance road safety, extend the life of tires, contribute to improved vehicle performance, and reduce overall operational costs among vehicle owners.

1.4 Objectives:

1. To check alignment for camber, toe, and caster of the wheel.
2. To measure tire pressure.
3. To provide early warnings for misalignment and unsafe tire conditions, ensuring optimal vehicle performance.
4. To create a guided tire misalignment detection system.

1.5 Organization of the Report:

This section explains arrangement of the report. It briefly tells about the contents which is discussed in each chapter of the report.

Chapter 1: Brief background/motivation/introduction, statement of problems, objectives, and organization of report.

Chapter 2: Literature review on Wheel Alignment, Tire Pressure Systems and Diagnostics Related to Sensors.

Chapter 3: Hardware components (sensors, ESP32) and software tools for implementation.

Chapter 4: System working principle, block diagrams, circuits, sensor integration, and firmware.

Chapter 5: Test result reports, readings, observations, and performance analysis.

1.6 Scope of the Proposed Work:

Scope 1: The object of the study is to produce a system capable of measuring the wheel alignment parameters by sensor measurements that are: camber, toe, and caster.

Scope 2: To propose an accurate tire pressure monitoring as an assessment of the health of tires in real-time.

Scope 3: To issue warning signals of misalignment or unsafe tires, improve the safety and performance of the vehicles.

Scope 4: To come up with guided tire misalignment system, which will guide the user in detecting the alignment problems in an efficient manner.

CHAPTER 2

CHAPTER 2

LITERATURE SURVEY

In this section, previous studies and currenty existing systems which are related to wheel alignment and tire health are discussed. Different types of methods such as workshop methods, sensor-based systems, IoT-based solutions, and vision-based systems are explained. The advantages and disadvantages of these methods are also mentioned. This study shows the problems like high cost, large size, and complex systems, and highlights the need for a small, low-cost, real-time embedded system.

2.1 Literature review:

Monitoring of wheel alignment and tire conditions of vehicles have become crucial fields of research in that their effect directly affects the safety of the vehicle, stability of handling stability, fuel consumption, and tire life. These are asymmetrical tyre treads, unstable steering, high rolling resistance and increased possibility of collisions aftereffects of incorrect wheel position, which include camber deviation, toe deviation and caster deviation. Most traditional wheel check systems are performed manually within a workshop and require massive equipment only experienced experts are able to cope with. Therefore, recent studies have directed the creation of automated, portable and real time monitoring. equipment capable of identifying the wrong alignment of wheels and other problems when the vehicle is in motion. The basic embedded systems, using microcontrollers and having simple sensors, to detect any anomaly on the. One of the principal examples of the earlier researches in this field is wheels. It was proposed that Arduino-based devices which combine temperature sensors, ultrasonic sensors and the simple ones to identify wheel misalignment, bear failure and abnormal tyre heating, environmental sensors should be installed. These systems would help to avoid tyre explosions and mechanical failures by warning the driver beforehand. Even though these methods proved the feasibility of integrated monitoring, they were over-dependent on proxy was restricted to measurements and traditional methods of sensing, which made them less accurate and responsive, especially in fast cars in which early intervention is vital. Car tracking systems based on IoT became the most popular in the wake of the development of wireless communication technology.

Various scholars proposed portable wheel tracking and tire pressure tracking solution that sent sensor data to smart phones or cloud computing platforms with Bluetooth, Wi-Fi or TCP/IP protocols.

These devices have the ability to perform preventive maintenance besides real time visualization and alarm to augment driver awareness. Significant IoT-based systems normally experience network dependency, smartphone incompatibility, signal issues. These drawbacks limit its use in the real-life driving environment where there is no necessity of constant communication possible. Optical and vision-based wheel alignment technologies were discovered in order to enhance detection accuracy by the researchers.

The parameter of alignment must be determined with high accuracy and hence the use of laser alignment, camera-based is one of the methods. 3D vision, target recognition and 3D vision have been explored. Vision-based systems are able to measure positioning of wheels and reference targets by depending on the spatial relation of wheels and reference targets camber, toe, and caster angle. These techniques, however, require a great deal of processing power and are very specific as regards to calibration and lighting conditions. It became possible through the use of artificial intelligence and machine learning to detect it to automatically find information and to make judgements.

Nonetheless, at the low-power embedded systems level, AI-based solutions will lead to more complexity and high processing higher prices, increased difficulties with real- time delivery. Automated detection structures have also been suggested using deep neural network and 3D vision technology to align the wheels and 30 days of maintenance. Inertial Measurement Units (IMUs), which are made up of accelerators and gyroscopes, are frequently used to determine the angular motion, orientation, and vibration characteristics of automobiles.

Experimental papers have demonstrated that IMU-based systems are useful in capturing the wheel dynamics and identifying the abnormal motion characteristics of the misalignment.

It has been noted that the precision of the system is greatly influenced by the correct subsystems, such as anti-lock brake systems, show that accurate inertial data significantly improves a system's performance and safety calibration of IMU sensors. Experiments on how IMU calibration affects automobile

Some researchers have offered mechanisms of reducing IMU drift and, through signal processing and sensor fusion. Such methods as zero-velocity update, frequency analysis with the usage of FFT, and better initial alignment have been is suggested to enhance the precision of inertial navigation. On the one hand, they enhance accuracy, whereas the performance is very sensitive to road conditions, the speed of the vehicle, etc.. on the other sensor quality and ambient noise. Bad terrain, vibrations and other external interference are capable of reducing performance particularly when using low-cost MEMS sensor. Another significant field of study is the application of IMU equipment in wheels.

The researchers placed IMUs on the suspension components or wheels and this made the results more because the IMUs were installed directly on the components or wheels direct measurements of behavior of either the wheels or contacts with the road. Such systems have been applied in vehicle localization, terrain mapping and with a wheel-based SLAM. The techniques also ease the hardware settings and minimize the use of various external sensors. It is also limited in the high-speed vehicles due to increased signal distortion. Hybrid sensor fusion techniques which incorporate wheel-speed sensors, IMUs, etc. need to be improved to enhance accuracy and robustness. The GNSS, LiDAR, and vision sensors have been investigated.

These systems utilize the advantages of the multiple sensors to compensate the disadvantages of the same, so as to allow them to perform effectively in diverse situations. Such systems cannot be used with small and low-cost embedded applications because they can be costly, intensive, and computationally intensive. Very sensitive inertial systems are under development such as fiber-optic gyroscopes which would achieve very high accuracy.

These systems reduce significantly drift and provide more efficient tracking of the path, but may not be used with consumer level automotive uses due to the expensive price, dimensions and power usage. Similarly, the research on the analysis of steering behavior using the IMU data points out the potentials of safety systems of vehicles include inertial sensors, however, the research is more focused on the study behavior of drivers than the condition of vehicles. Most of the literature, as a rule, illustrates that accuracy, complexity and practicality are at a trade- off in the present wheel alignment and car surveillance systems. Both of the vision and AI based systems are extremely specific but costly and have low calculation capabilities the naive sensor- based systems are not quite robust and accurate.

The solutions following the IoT make them more accessible, but they are highly dependent on the quality of communication. Inertial sensing and sensor fusion techniques show high potentiality yet, they need to be calibrated carefully. The above findings indicate that there is a research gap in this field of compact, cost effective and real-time wheel misalignment detector system which is based on the combination of inertial sensing and auxiliary sensors but low. This is highly demanded by having an embedded solution that will enable the early detection of alignment problems even in real condition of driving without the use of sophisticated vision systems and without constant network connectivity. The current work fills this gap by paying attention to a practical sensor-based solution to incorporate inertial, pressure, and wireless measurements to improve vehicle safety and efficiency in terms of maintenance.

The other significant point that has been brought to light in the literature is the increasing focus on fault detection at times of wear and tear, instead of the diagnosis of failures in automotive systems. It has been suggested by a number of researchers that standard alignment checks are generally undertaken after observable symptoms like uneven wear on tires or steering pull have been observed which means that some quantifiable damage will already be done. Research points out that by continuously or periodically monitoring wheel parameter during vehicle operation, proactive maintenance can be done, long-term repair cost can be minimized and the overall vehicle dependability is enhanced.

This reactive to preventative monitoring has prompted the requirement that this be frequently operated by humans. Moreover, the former study demonstrates the importance of tyre temperature and pressure as the supplementary ones indicators of the well-being of the wheel alignment.

In order to enhance the specificity of problem diagnostics, there are studies that combine the detection systems of alignment and pressure and temperature sensors. Nevertheless, most of the existing approaches consider these factors separately and are not correlated them with divergent alignments. This absence of analysis in one piece renders them incapable of delivering an extensive study of wheel condition, which offers an opportunity to come up with systems that can integrate multiple sensor inputs to offer a more precise output assessment. The usage of wireless communication forms an important part of the current vehicle monitoring system that provides information recording, distance diagnostics, and real-time alerts.

Wireless technologies on short ranges have been proven to be specially handy in embedded automotive systems because of their low power consumption and integration. Nonetheless, other issues such as delay in data, interference and signal reliability are also reported. To make sure that important messages are delivered to the driver within their immediate environment in the eventuality of wireless communication breakdown, there are research studies that have proposed the development of systems that can still work even in the event of sporadic connectivity.

Regarding embedded automotive solutions, a number of specialists have mentioned the significance of low power and low complexity of computing. The high-end processors and complicated algorithms are cheaper and more efficient. Lightweight filtering is used in real-time applications especially in portable and handheld diagnostic instruments. Thresholds based decision logic, simplified calibration processes are gaining popularity. Finally, but not the least, the literature also clearly highlights the importance of sound calibration and validation processes to provide the same level of system performance when used in a variety of vehicles and operating conditions.

These are because of differences in suspension geometry of vehicles, loading conditions, road geometry and sensor placement factors highly influence the accuracy of measurements. Other authors identify that false positives or false missed detection can be caused by faulty calibration, which can undercut confidence of automated monitoring systems by the user. Thus, the concept of flexible calibration and comparison methods in which the reference would serve to compare the variables would have been made possible systems to adapt to various vehicle environments without affecting the accuracy of detection is associated with modern trends of research.

2.2 Overview of literature survey:

Literature review overview Findings: It is required that wheel misalignment should be attended to maintain the safety of the vehicle, enhance its performance, and reduce the maintenance cost. The literature identifies the progress made to the vision-based, sensor-based, and Internet of Things-based systems. The summary contains the most important strategies, trends, and challenges in the industry to the extent that expediency is involved and cost- effectiveness. Standard accelerators detect tilt, vibrations and other such like aspects using standard steering sensors and pressure sensors.

Such systems tend to struggle being precise in dynamic or due to sensor noise and deficiency of pliability IoT-based Solutions Real-time Tracking of alignment data through cloud or mobile. Applications are made possible through IoT integration. These systems are more functional and convenient since the user can access alerts and diagnostics within a short time. However, smart phones and wireless networks have problems with accessibility particularly in remote locations, places or locations where poor signals are experienced.

Vision-Based and AI-Driven Systems: Advanced systems with the use of 3D vision technologies or AI algorithms imply a greater level of accuracy and a more complete analysis of the parameters of alignment. Such approaches are promising because of their accuracy, but have drawbacks of requiring excessive computing ability, increased cost and limited ability to execute in real-time and with limited resources. Energy Efficiency and Practicality: More recent attempts are aimed at making detection systems in alignment energy-efficient and practical in a variety of conditions. Some of the techniques include sensor dual wake-up and simplified frameworks that are aimed at maximizing performance and reducing complexity. Nevertheless, a compromise between cost, accuracy, and usability is not an easy task.

CHAPTER 3

CHAPTER 3

HARDWARE AND SOFTWARE REQUIREMENTS

The chapter describes the hardware and software that were involved in the wheel alignment detection system. It explains the sensors, microcontroller units and display that will be used in the system. These elements are integrated in order to track wheel alignment issues in real time.

3.1 Hardware Requirements:

3.1.1 AS5600 Sensor (Steering angle):

AS5600 is a high-precision, non-contact magnetic angle sensor that can be applied within a wheel-alignment monitoring system to properly track the rotation and position of wheels. Table 3.1 demonstrates the AS5600 sensor specifications. Figure 3.1 represents the AS5600 sensor, which is proposed to be utilized in the given structure of the work and provides an opportunity to control wheel alignment in real time in order to avoid inequality in tyre wear and to increase fuel efficiency.

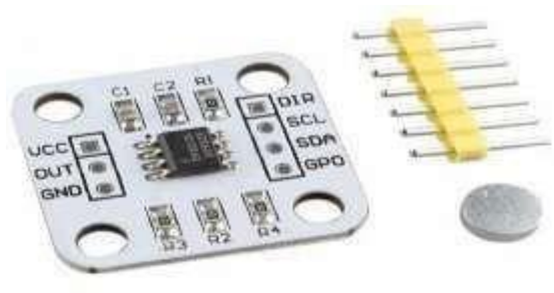


Figure 3.1 AS5600 Sensor

AS5600 Module Features:

- Contactless angle measurement.
- Simple user-programmable start and stop positions over the I²C interface.
- Maximum angle programmable from 18° up to 360°.
- 12-bit DAC output resolution.
- Analog output radiometric to VDD or PWM-encoded digital output.
- Automatic entry into low power mode.

Table 3.1 AS5600 Sensor Specification

Parameter	Specification
Type	Magnetic Rotary Position Sensor
Resolution	12-bit (0–4096 positions)
Output Interface	I²C / PWM
Operating Voltage	3.3 – 5 V
Accuracy	High angular precision

Importance of AS5600 Sensor in Wheel alignment monitoring:

- Non-contact measurement: The AS5600 detects position by use of a magnetic field, without physical contact with the wheel or axle. This helps minimize wear and tear on the sensor, as well as provide proper readings with time.

- Compact size: - The AS5600 comes in a small business package; hence, it can be easily incorporated into wheel alignment systems.
- Prevent uneven tire wear: - By detecting and correcting misalignment, the system can help
- prevent premature and uneven tire wear, extending tire life.
- Improve fuel efficiency: - Misaligned wheels can cause increased rolling resistance, leading to reduced fuel efficiency. Monitoring alignment can help optimize fuel consumption.
- Enhance vehicle safety: - Misalignment can affect handling and stability, potentially compromising safety. Monitoring alignment can help ensure safe driving conditions.

3.1.2 ADXL345:

The ADXL345 is a compact, small, and low-power 3-axis digital accelerometer which has a wide range of applications in motion and tilt sensing, and orientation measurements. This system requires the sensor to measure the fixed tilt of the wheels; this is necessary to obtain the wheel alignment parameters such as the camber and the caster of the wheel assembly under the fixed test regime of the vehicle when in a stationary state. Table 3.2 shows the specifications of the ADXL345 sensor.

Figure 3.2 ADXL345

Role of ADXL345:

The ADXL345 gets attached to the wheel configuration or alignment frame to measure the changes in the inclination at rest.

Its major duties will involve:

- Recording the tilt of wheels during wheel inspection.
- Giving crude acceleration data to the ESP32 controller.
- Camber and caster real-time computations.
- Providing the opportunity to detect abnormal angular deviations based on preset thresholds.
- Since measurements are done at stationary conditions the sensor provides stable noisy free results without involving methods of complex filtering or sensor fusion.

Table 3.2 ADXL345 Specification

Parameter	**Specification**
Type	3-Axis MEMS Digital Accelerometer
Measurement Range	±2 g, ±4 g, ±8 g, ±16 g
Resolution	13-bit
Supply Voltage	2.0 – 3.6 V
Output	Digital via I²C / SPI
Current Consumption	~140 µA

Importance of ADXL345 Sensor in Wheel alignment monitoring:

- Precise Tilt Detection: By maintaining precise monitoring of minor wheel inclinations, camber and caster deviation can be detected early.
- Stationary Suitability: The sensor is the best in a stagnant environment where angle measurement is best done by gravity.
- Low Power Consumption: Ideal where there is constant use with a rechargeable battery.
- Streamlined Integration: Digital communication and compatibility with ESP32 microcontroller are features that give the board an easy time in hardware interfacing.
- Cost-Effective Solution: Offers a professional way of sensing with a relatively low expense of industrial level alignment equipment.

3.1.3 Resistors (10 kΩ and 15 kΩ):

The resistors are passive electrical elements, which are used to control the amount of current and voltage in the electronic system; Figure 3.3 shows 10 kΩ and 15 kΩ resistors used in the system to condition signals, put the circuit into pull-up and pull-down modes, create voltage divider networks, and provide a biasing reference when the sensor is interfaced with the microcontroller. Table 3.3 shows the properties of the resistors (10 kΩ and 15 kΩ).

Figure 3.3 Resistors (10 kΩ and 15 kΩ)

Resistors (10 kΩ and 15 kΩ) Features:

- Give constant and steady values of resistance values of the circuits.
- Easy and very dependable passive parts.
- Cheap and readily accessible.
- Applicable in voltage divider and biasing circuits.
- Supersets with low-power microcontroller-based designs.
- Minimal power dissipation.

Table 3.3 Resistors (10 kΩ and 15 kΩ) Specification

Parameter	Specification
Type	Fixed resistor
Resistance Values	10 kΩ and 15 kΩ
Tolerance	±5%
Power Rating	0.25 W (¼ watt)
Package	Axial lead

Resistance of Resistors in Wheel alignment monitoring:

- The latter are used to shape the input signal with voltage divider circuits.
- Give pull-up and pull-down capabilities to stabilize digital logic signals.
- Aid in shielding sensors and ESP32 microcontroller against large currents.
- Assure steady reference levels needed in selecting sensor data in a uniform manner.
- Enhance total reliability and accuracy of measurements of circuits.
- Provision of secure and productive functionality of the embedded hardware system.

3.1.4 Jumper Wire:

Jumper wires are thin electrical conductors, which are used to create temporary or semi-permanent connections between development boards and sensor modules. Figure 3.4 shows the jumper wires that can be used to connect the ESP32 development board to the ADXL345 sensor, the AS5600 sensor, the MPX5700AP sensor, and the OLED display. They enable quick assembly of a circuit, easy debugging, and quick modification without the use of a soldering iron.

Figure 3.4 Jumper Wires

Jumper Wire Features:

- Provide fast and non-soldering electrical connections.
- Flexible designed multi-configurable tests.
- Colour coding- it helps in sorting out the wiring and reducing mistakes made in connection sets.
- Quickly test faults in electrical circuits and develop hardware.
- Breadboard compatible and integration compatible.
- Jumper Wires Briefing in the Project.

- Constructed into what will fit into the ESP32 controller and have all the sensors and modules connected in a dependable manner.
- Make prototyping and debugging of the system-hardware efficient.

3.1.5 TP4056 Charging module:

The TP4056 component is a linear Li-ion/Li-polymer battery charger controller, which is used to provide safety when charging single-cell rechargeable batteries. Table 3.5 shows the TP4056 charging module. Figure 3.5 illustrates the TP4056 charging module that will be adopted to regulate and control the process of charging the battery in order to obtain maximum power control, as well as to avoid overcharging of the lithium battery resulting in damage.

Figure 3.5 TP4056 Charging module

TP4056 Charging module Features:

- Battery-charging control which is safe and reliable.
- Automatic battery protection circuitry.
- Basic USB based input charging.
- Small number of components and small size.
- Simple interconnection with mobile systems.
- Relevance of TP4056 to the Project.
- Provides secure charging of the lithium battery that is used to charge the system.

Table 3.5 TP4056 Charging module specifications

Parameter	**Specification**
Type	Li-ion Battery Charging Controller
Input Voltage	5 V (USB)
Charging Current	Up to 1 A
Charging Method	Constant Current / Constant Voltage
Protection	Over-charge & short-circuit protection
Indicators	Charging & full LEDs

3.1.6 Lithium-ion battery:

The project utilizes the lithium-ion battery as the main portable energy source due to its small size and large energy density. Table 3.6 shows the lithium-ion battery specifications. Figure 3.6 illustrates the lithium-ion battery which enables the ESP32 controller, sensors, and display to be powered without an external power adapter.

Figure 3.6 Lithium-ion battery

Lithium-ion battery Features:

- Higher power density to achieve a long operating time.
- Lightweight and portable.
- Low self-discharge rate.
- Batteries that are rechargeable and environmentally friendly.
- Offers constant voltage.
- Value of Lithium Battery in the Project.

Table 3.6 Lithium-ion battery specifications

Parameter	Specification
Nominal Voltage	3.7 V
Fully Charged Voltage	4.2 V
Capacity	1000–2000 mAh
Type	Rechargeable Li-ion
Cycle Life	300–500 cycles

3.1.7 MT3608 Booster Module:

The MT3608 is a DC-to-DC step-up (boost) converter which allows the low input voltage to be increased to a higher regulated output voltage. In Table 3.7, the specifications of the MT3608 booster module are shown. Figure 3.7 shows the booster module, which increases the battery voltage to 3.7 V before the required voltage is supplied to the OLED display and other peripherals in the system.

Fig 3.7 MT3608 Booster Module

Table 3.7 MT3608 Booster Module specifications

Parameter	Specification
Input Voltage	2 – 24 V
Output Voltage	Adjustable up to 28 V
Current Capacity	Up to 2 A peak
Efficiency	Up to 93%
Control	On-board trimmer potentiometer

MT3608 Booster Module Features:

- Good voltage boosting ability.
- Adjustable output voltage range.
- Lightweight and small construction.
- Conversion of power is very efficient.
- Simple plug and play interfacing.
- Applicability of MT3608 to the Project.
- Brings proper amount of voltage to high-voltage peripherals.
- Enhances continuous operation of the system when there is battery depletion.

3.1.8 OLED display:

To present real-time wheel alignment parameters and tyre pressure readings to the user, the OLED display module is used. Figure 3.8 shows the OLED display, which forms the main user interface. Its display is particularly high-contrast, with excellent viewability and low power consumption due to the self-illuminating pixel technology, thereby simplifying the interpretation of sensor data when performing wheel alignment in a stationary position and tyre health checks. Table 3.8 shows the specifications of the OLED display.

Fig 3.8 OLED Display

Features of OLED Display:

- Bright display and high contrast so that it can be seen very easily in various lighting conditions.
- No back light is necessary since it has self-illuminating pixels which bring a reduction in power whoosh.
- Small size and lightweight construction, recommended to be used in portable embedded systems.
- Services simple text and plain graphical point of presentation of sensor values.

Table 3.8 OLED Display specifications

Parameter	Specification
Display Type	OLED Graphical Display
Resolution	128 × 64 pixels
Interface	I²C
Operating Voltage	3.3 – 5 V
Display Size	0.96 inch
Power	Low consumption, no backlight

Importance of OLED Display in Wheel alignment monitoring:

- Plays the central role in the form of visualisation of the wheel alignment angle and tyre pressure values.
- Helps enables the user to easily detect when there is anomaly in the alignment or unsafe tyre pressure without relying on external apparatus.
- Gives high contrast and is easily readable hence data is well interpreted.
- Draws minimal power and it can therefore be operated by batteries.

3.1.9 Foam Cardboard:

Table 3.9 shows the foam cardboard specifications, and Figure 3.9 shows the foam cardboard base that was utilized in the process of accurately positioning the sensors during the stationary alignment test, and that was used to provide electrical insulation and easy fabrication.

Figure 3.9 Foam Cardboard

Table 3.9 specifications

Parameter	Specification
Type	Lightweight mounting base
Structure	Foam core with paper layers
Properties	Electrically insulating
Usage	Sensor and module support frame

3.1.10 Double-Sided Foam Tape:

The mounting components and the lightweight sensors are bonded onto the foam cardboard base with the assistance of non-drilled foam tape, and no permanent adhesive bonding was utilized. Figure 3.10 indicates the manner in which the sensor modules and display unit are attached with the aid of double-sided foam tape.

This ensures that the ADXL345 accelerometer, AS5600 angle sensor, and the OLED display module are well secured, which is essential for effective data collection. Table 3.10 indicates the properties of the double- sided foam tape used, and the tape can also be subjected to easy repositioning of the parts when performing calibration and verification of the experiment.

Figure 3.10 Double-Sided Foam Tape

Table 3.10 Double-Sided Foam Tape specifications

Parameter	Specification
Type	Adhesive mounting foam tape
Bond Strength	High
Vibration Absorption	Yes
Usage	Temporary component mounting

3.1.11 ESP32 development board:

ESP32 Development Board is a microcontroller board that is an all-purpose and powerful microcontroller that is extensively applied in the IoT and embedded automation systems. It is designed based on a dual-core Tensilica LX6 processor that has high processing speed and multitasking efficiency. The ESP32 Development Board is the main controller in the system, as shown in figure 3.11. The board has inbuilt 2.4 GHz Wi-Fi and Bluetooth (Classic and BLE) to allow wireless communication without the use of an external module. These features render the ESP32 to be an appropriate board in near real-time data exchange, remote sensing, and sustained wireless management that is demanded in embedded diagnostics systems.

Figure 3.11 ESP32 Development board

Importance of ESP32 development board in Wheel alignment monitor:

- Acts as the Main Control Unit.
- Wireless Data Transmission Enables.
- Supports Multiple Sensor Interfaces.
- Properly to work with batteries.
- Powerful Processing to give accurate monitoring.
- Small and Fit in any Processor environment.

3.1.12 Silicone Tube:

The air pressure hose that is in use to transfer the air pressure between the tyre valve assembly and the pressure sensor inlet is silicone tubing. The silicone tubing connection to be used as a tyre pressure measurement device is illustrated in figure 3.12. The choice of silicone tubing is based on its strength, flexibility and ability not to bend or crack.Table 3.12 presents the silicon tube characteristics. The tubing provides an airtight closure along the entire length of the pressure line to eliminate leakage, which would cause errors in accuracy of sensors. Its resistance to temperature changes and mechanical stress ensures the transmission of pressure and durability of its use over time when experimental research is being repeated several times.

Table 3.12 Silicone Tube specifications

Parameter	Specification
Type	Flexible pneumatic tubing
Material	Silicone rubber
Properties	Airtight, flexible, durable
Usage	Connecting tyre to sensor

Figure 3.12 Silicone Tube

3.1.13 Zip Ties:

The wire cables, tubing and sensor leads are affixed to the foam cardboard base or support structure using zip ties. The zip ties utilized in the setup can be seen in figure 3.13. It is important to manage cables properly so as to prevent loose connections. Table 3.13 gives the zip ties specifications. Zip ties are used to maintain a system that is neat and that is mechanically stable and the advantage is that the strain on the electrical connections is reduced and also the safety and reliability of the entire experiment is enhanced.

Table 3.13 Zip ties specifications

Parameter	**Specification**
Type	Cable fastening ties
Material	Nylon
Length	10–20 cm
Usage	Wiring and tubing management

Figure 3.13 Zip ties

3.1.14 Perfboard:

Perfboard is a solderable prototyping board, which can be used to take a permanent electrical circuit. It is shown on figure 3.14 in which the electronic components will be assembled and connected on the perfboard in the system. The components that are soldered to the perfboard in this project are resistors, voltage regulators, connectors and jumper wire interfaces that establish fixed electrical connections between modules. Table 3.14 shows the specification of the perfboard.

Table 3.14 Perfboard specifications

Parameter	Specification
Type	Solderable prototyping board
Material	FR-4 or phenolic laminate
Hole Pitch	2.54 mm
Pad Type	Copper isolated pads
Usage	Permanent circuit assembly

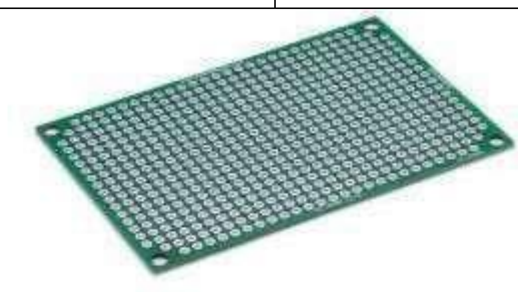


Figure 3.14 Perfboard

3.1.15 Mpx5700ap pressure:

The MPX5700AP is a piezoresistive silicon pressure sensor with high accuracy, applicable in gauging air pressure. Figure 3.15 depicts the tyre pressure sensor in the system, which is the MPX5700AP. The specifications of the MPX5700AP pressure sensor are presented in Table 3.15. The sensor is used to measure tyre pressure in a stationary state in the Smart Wheel Misalignment and Tire Health Monitoring System.

Figure 3.15 MPX5700AP pressure

Mpx5700ap pressure Features:

- Wide pressure measurement range suitable for automotive tyres.
- Linear voltage output simplifies microcontroller interfacing.
- Built-in temperature compensation for stable readings.
- Long-term reliability and consistent performance.
- Compact and robust construction.

Table 3.15 MPX5700AP pressure specifications

Parameter	Specification
Type	Analog Pressure Sensor
Measurement Range	0 – 700 kPa
Supply Voltage	5 V
Output Type	Analog voltage
Accuracy	High sensitivity, calibrated output

Importance of MPX5700AP pressure:

- Used to measure tyre air pressure accurately during stationary testing.
- Provides real-time pressure data to the ESP32 controller for analysis.
- Enables detection of under-inflated or over-inflated tyre conditions.
- Enhances overall system capability by supporting tire-health monitoring along with alignment detection.
- Supports early identification of unsafe driving conditions.
- Contributes to improved vehicle performance and tyre lifespan through accurate monitoring.

3.1.16 Switch:

A switch commonly describes a simple electromechanical device that is employed to open or close an electrical connection that carries electrical power in a circuit. The switch in the system that is used to control power is depicted in figure 3.16. Table 3.16 demonstrates the switch specifications.Switches will be applied to safely turn on and off the sensor nodes and the receiver unit to enable the system to save battery power when the system is not in operation and to provide control over the start-up of each module.

Figure 3.16 Switch

Switch Features:

- Simple and reliable manual power control.
- Compact size suitable for portable devices.
- Easy integration with low-voltage DC circuits.
- Durable construction for repeated operation.
- No external driving circuitry required.

Importance:

- Used to control the main power supply to sensor nodes and the receiver unit.
- Helps to prevent unnecessary battery discharge when the system is idle.
- Enables safe startup and shutdown of electronic modules.
- Provides user convenience for device operation during testing and demonstrations.

3.1.17 ESP32-C3:

ESP32-C3 has an in-built Wi-Fi (2.4 GHz) and Bluetooth Low Energy (BLE 5.0) connection. The ESP32-C3 microcontroller module installed in the system is illustrated in Fig. 3.17. Besides that, the device also embraces superior security measures that include hardware-based encryption and secure booting, which guarantee secure and dependable information transmission in real-time embedded systems.

Figure 3.17 ESP32-C3

3.2 Software Requirements:

The integrated development environment software platform was the Arduino Integrated Development Environment (IDE) and was employed to write, compile, and transfer embedded code into ESP32 microcontroller. Its interface is easy to use in the following aspects; supporting C/C++ based firmware, controlling libraries, configuring the board, and serial monitoring. The device that serves as a central component in the case of Smart Wheel Misalignment and Tire Health Monitoring System is the Arduino IDE, which will be used to connect with the Adamic accelerator ADXL345, the angel sensor AS5600, the pressure sensor MPX5700AP, and the OLED monitor. The specifications of Arduino IDE are depicted in Table 3.18. It is simple and has a significant library base that makes it possible to retrieve sensor data with maximum efficiency and run the computations to determine the parameters of alignment and visualize the findings. The Arduino IDE (software) is illustrated in figure 3.18.

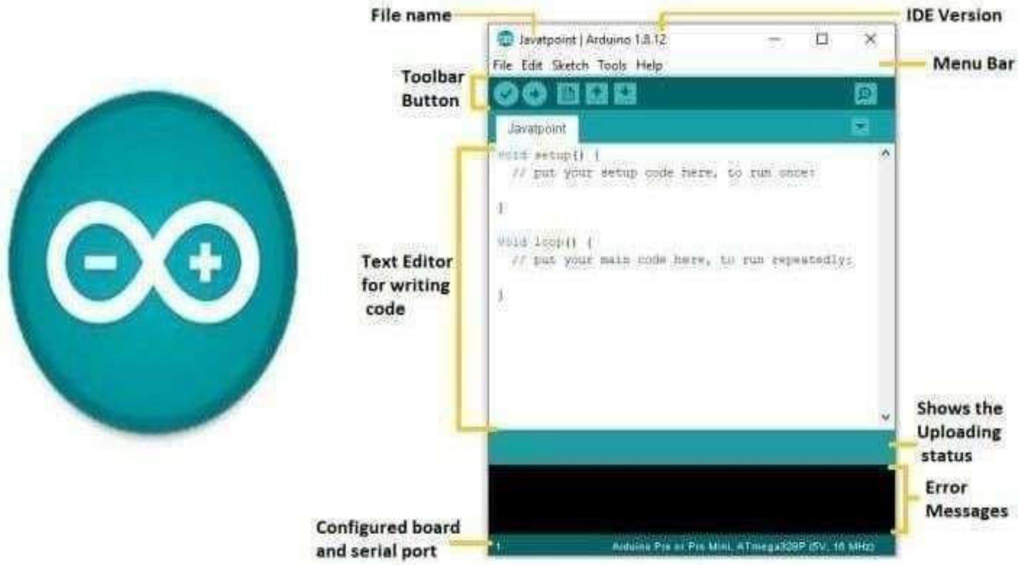


Figure 3.18 Arduino IDE (software)

Application of Arduino IDE:

The program, which is to execute all the activity of the system, is coded and entered in the Arduino IDE. It allows:

- The ESP32 microcontroller to interrelate with sensors to acquire raw data on the ADXL345 accelerator to be used to determine the tilt angles of the wheel.
- The MPX5700AP sensor that measures the health of the tyres picks up the pressure.
- With stationary systems and data crunching, camber, caster and toe parameters are approximated.
- The contact with the OLED screen which will provide the user with the measured values.

Table 3.18 Arduino IDE Specifications

Parameter	Specification
Type	C/C++
Language Support	Embedded software development IDE
Supported Boards	ESP32, Arduino
Tools	Compiler, uploader, serial monitor
Platform	Windows, Linux, macOS
Licensing	Open Source

Significance of Arduino IDE in Wheel alignment monitoring:

 The Arduino IDE (Integrated Development Environment) plays a crucial role in wheel alignment detection projects by serving as the platform to write, compile, and upload code to an Arduino microcontroller.

 Arduino IDE enables one to code in order to receive their data and analyse it to identify the wheel alignment with control sensors discrepancies. The Arduino can be configured to keep track of sensor readings and make decisions when some wheel may be out of line.

 The IDE has serial monitoring and debugging tools, which are beneficial in checking precision and reactiveness of the wheel alignment system.

 Processing of real-time can be done using data written in the Arduino IDE data to determine angles of alignment, distances, and deviations to the expected sensor data values. The processed data may be used to identify the alignment of the wheel right or whether corrections are required.

CHAPTER 4

CHAPTER 4

METHODOLOGY AND IMPLEMENTATION

In this section, the description of the functioning of the proposed system and the way it is constructed are provided. It explains the connections between the sensors and the ESP32 microcontroller and wireless transmission of the measured data. It is also described how it detects wheel misalignment and tire defects. It entails block diagrams, circuit design, and firmware development as well as step-by-step implementation to demonstrate how system is used to offer real-time monitoring and alert messages.

4.1 Methodology:

The given Wheel Misalignment Detection System is a proposed system that is a distributed sensor network, assessing wheel alignment and tyre health in real time. The design, deployment, wireless communication, and data processing of three autonomous sensing nodes and one central receiver unit form the methodology. Figure 4.1 presents the block diagram of the methodology. Each of the stages of development is outlined below.

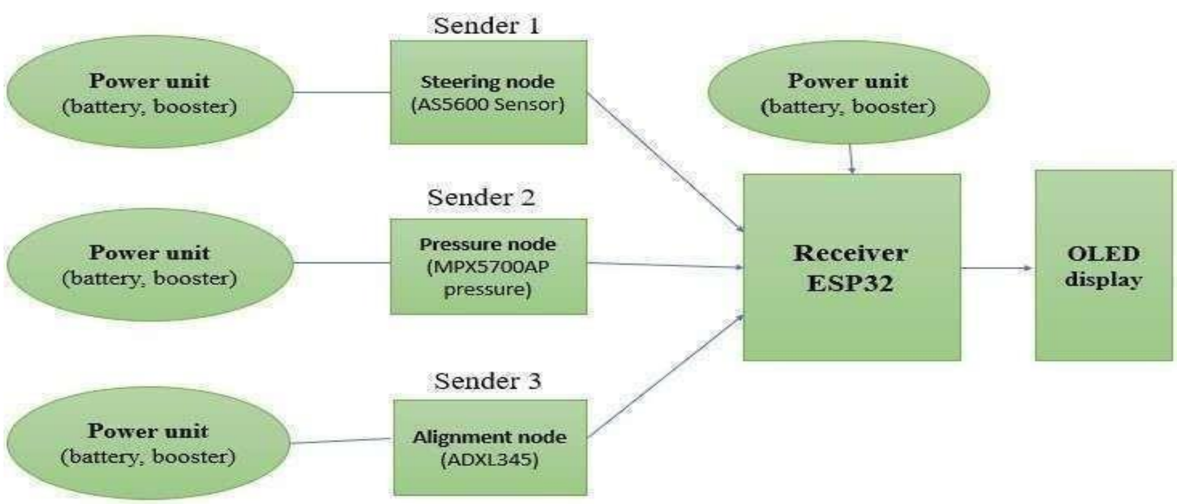


Figure 4.1 Block diagram of Methodology

4.1.1 ADXL345 Sender Module:

The ADXL345 sender is an instrument that measures wheel tilt and vibration parameter in terms of camber and caster angle. ADXL345 accelerator has three axes of acceleration and translates the physical tilt of the wheel into digital form. The sensor is connected to ESP32-C3 microcontroller that interprets the readings of acceleration and packages them to be sent wirelessly. The module is powered by a Li-ion battery, which can be controlled with a TP4056 charging module and an MT3608 boost converter, to make sure that the voltage remains stable. Figure 4.2 demonstrates that the processed sensor data are sent to the receiver unit to be further analyzed so that the current conditions of wheel alignment could be monitored in real-time.

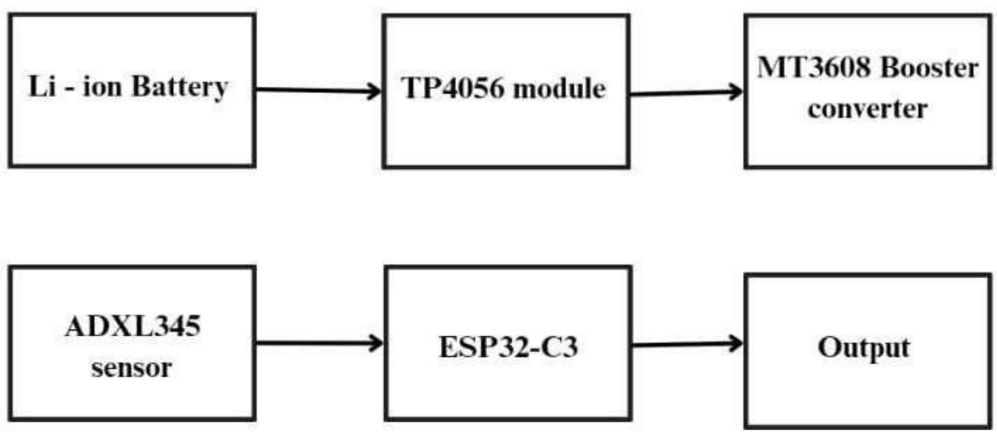


Figure 4.2 ADXL345 Sender Module

4.1.2 AS5600 Sender Module:

AS5600 sender module has the job of measuring the steering angle, which is utilized to estimate toe alignment changes. The AS5600 magnetic encoder is a magnetic measuring device that determines the angular position variation with rotation of a magnet on the steering mechanism. This information is subsequently sent wirelessly to the receiver module to be displayed and interpreted as shown in Figure 4.3 as part of proper toe angle evaluation.

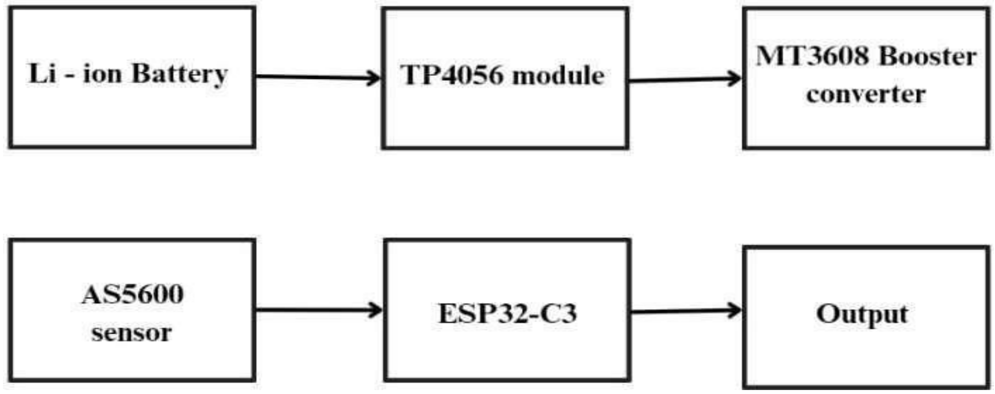


Figure 4.3 AS5600 Sender Module

4.1.3 MPX5700AP Sender Module:

The MPX5700AP sender unit is in charge of detecting the pressure in the tyre and measuring it into an electrical signal which can be processed. The MPX5700AP pressure sensor is a sensor which detects a change in air pressure and generates an analog voltage according to the pressure exerted. The conditioned signal is fed to the ESP32-C3 microcontroller where it is digitized and sent further. To supply the power to the module, a rechargeable Li-ion battery is used and controlled by TP4056 charging module to provide safe charging and battery protection. Figure 4.4 demonstrates the supply of a constant voltage to the sensor and controller to ensure the sensor and controller operate reliably under a changing set of operating conditions by the use of a boost converter of type, MT3608, to supply a constant voltage to the sensor and controller.

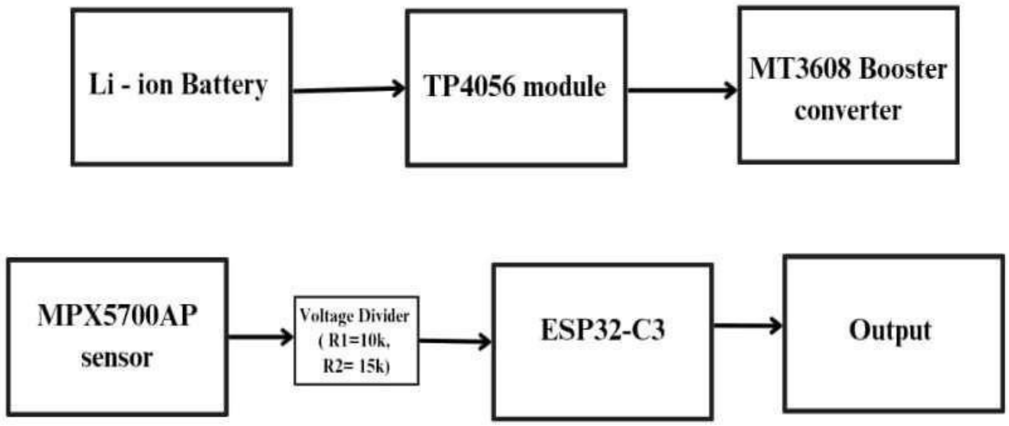


Figure 4.4 MPX5700AP Sender Module

4.1.4 ESP32 Receiver Module:

The sensor data receiver is centered on the ESP32 receiver which is the receiving, processing, and displaying sensor data receiver. The ESP32 microcontroller works with the sensors and data collected by them (such as the ADXL345 and MPX5700AP) to provide parameters of wheel alignment and tyre pressure. The processed information is then put on an OLED screen to be easily interpreted by the user. The receiver module connects to power via a Li-ion battery, and is charged by a TP4056 and voltage regulated using a MT3608 boost converter. This design is effective in terms of power control and constant functioning. The general communication between the input sensors, ESP32 processing unit and the display interface is shown in Figure 4.5 and it denotes the flow of data in the receiver module as shown in the figure.

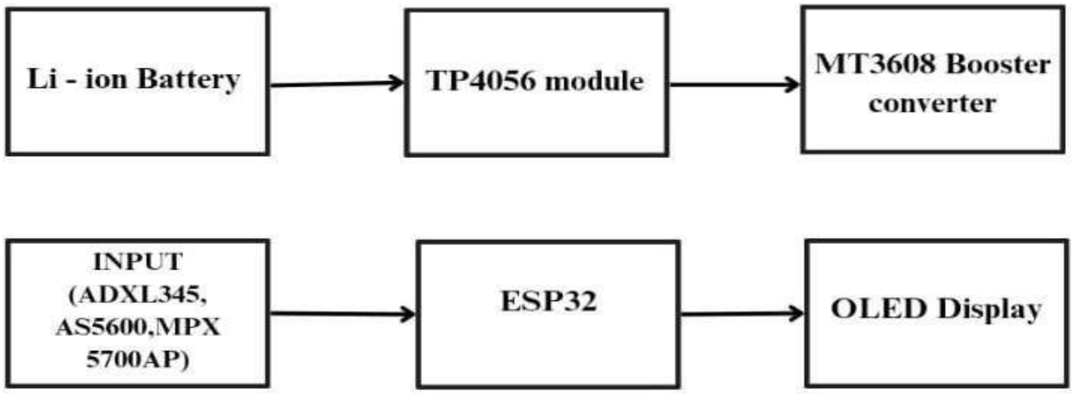


Figure 4.5 ESP32 Receiver Module

4.1.5 System Architecture Overview:

The system uses a distributed wireless setup with three sensor nodes mounted on the vehicle wheels and steering. These communicate via ESP-NOW with a central handheld receiver ESP32

➢ Sensor Nodes (3 units):

- ADXL node for camber and caster angle detection.
- Steering node for steering angle measurement using AS5600.
- Pressure node for tyre pressure measurement using MPX4115AP.

➢ Main Receiver Unit (Handheld device):

- ESP32-Dev module with OLED display for live output visualization.
- Each All sensor nodes transmit data wirelessly to the receiver using the ESP-NOW protocol.The receiver processes multi-sensor data to estimate camber,caster, toe deviation, and tyre pressure condition.

4.1.6 Sensor Node Design:

All the sensor nodes consist of a small battery-powered board containing an ESP32-C3 microcontroller, a dedicated sensor module, an MT3608 DC–DC booster to regulate voltage, and a 3.7 V Li-ion battery with a TP4056 charging module, with the battery connected to the TP4056 and then to the MT3608 and ESP32-C3 + sensor, following one uniform power flow, which includes Battery → TP4056 → MT3608 → ESP32-C3 + Sensor.

ADXL Node – Wheel Orientation Detection:

- Sensor: ADXL345 accelerometer.
- Placement: Front-left wheel knuckle using foam + double-tape damping to minimize vibration noise.
- Function: Reads 3-axis acceleration for inclination analysis.
- Processing:
  - X–Z plane → Camber angle.
  - Y–Z plane → Caster angle.
- Wireless Task:
  - Packs processed values (X, Y, Z float data) and sends via ESP-NOW.

Steering Node – Toe Angle Estimation:

- Sensor: AS5600 magnetic rotary encoder.
- Placement: Coupled to steering shaft using magnet-to-sensor alignment.
- Function: Measures rotation in 0–4096 resolution.
- Processing: Converts raw position to steering angle in degrees.
- Wireless Task: Sends 16-bit steering value via ESP-NOW.

Pressure Node – Tyre Pressure Monitoring:

- Sensor: MPX4115AP analog pressure sensor.
- Connection: Sensor output is fed through a voltage divider (15 kΩ + 10 Ω series) to ESP32- C3 ADC.
- Placement: Connected to tyre air valve using silicone tube and brass T-adapter.
- Processing: Converts raw position to steering angle in degrees.
- Wireless Task: Sends 16-bit steering value via ESP-NOW.

4.1.7 Main Receiver Unit:

The receiver unit is a diagnostic unit on a handheld scale, used to perform alignments, placed next to the car to perform inspections, and consisting of an ESP32-Dev microcontroller, a 0.96-inch OLED display with an I2C interface, a blue LED indicating that packets have been successfully received, a Li-ion battery as a power source with a TP4056 charge controller, and an MT3608 boost conversion stage. It tracks the reception of all three sensor nodes' ESP-NOW packets and groups them according to the size of the packet, and when it is given a sensor data set, it computes the following:

It averages the accuracy of the algorithm over 100 samples and estimates the sensor node signal.

Parameter	Computation Method
Camber	atan2(X, Z)
Caster	atan2(Y, Z)
Toe	Steering deviation × toe-calibration factor
Pressure	ADC → Voltage → kPa → PSI

Finally, a decision layer compares calculated values against defined thresholds to determine:

- Aligned
- Alignment needed
- Pressure Issue

The result is displayed on the OLED along with individual parameter values and live status (OK / Slight / Bad / Out / In / Low / High).

4.1.8 Deployment Procedure:

Step 1: Attach sensor nodes firmly near their corresponding measurement points. Step 2: Power on all sensor nodes and the handheld receiver unit.

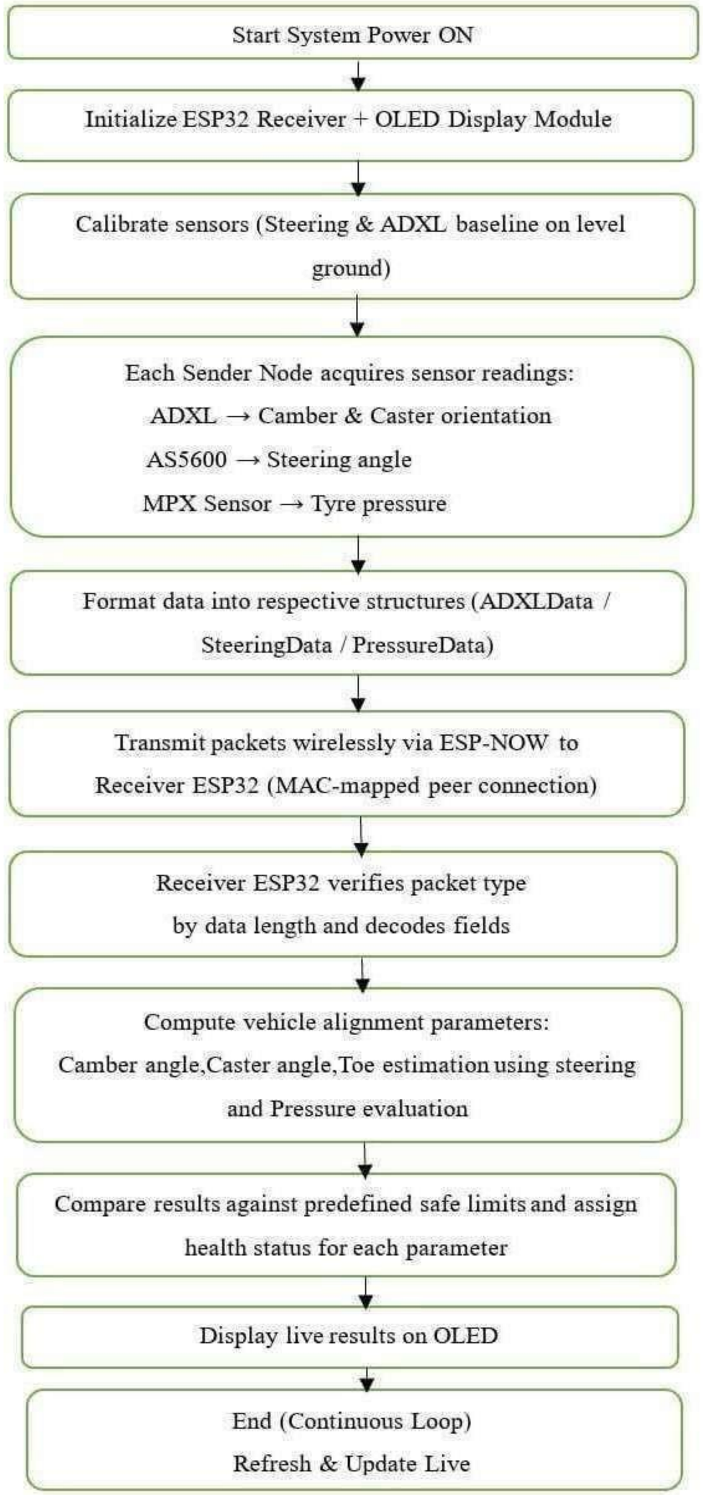
Step 3: The nodes automatically connect to the receiver over ESP-NOW.

Step 4: Live wheel alignment and tyre pressure values appear on the OLED display. Step 5: Any deviation instantly updates the status and recommendation on the screen.

4.2 Implementation:

The Wheel Misalignment Detection System was adopted as a distributed wireless sensor network which consists of several autonomous sensing nodes and one handheld receiver unit.The sensing nodes are designed in a manner that it uses an ESP32 based controller with special sensors to measure each of the wheel parameters: camber, caster, steering angle and tyre pressure. Sensor data is locally interpreted, packaged into formatted packets and sent wirelessly to the receiver unit with the ESP-NOW communication protocol.The receiver unit receives the packet of data, identifies and decodes it, and finishes the data fusion process, and calculates the parameters of the wheel alignment by using predetermined estimation algorithms.

The results obtained are then compared to the safe threshold limits so as to ascertain the alignment and the condition of the tyres. Lastly, the real-time values of the alignment and health status are also presented on an OLED interface, and the system is powered continuously to offer real-time updates during stationary inspection.


Start System Power ON
Initialize ESP32 Receiver + OLED Display Module
Calibrate sensors (Steering & ADXL baseline on level ground)
Each Sender Node acquires sensor readings:
ADXL → Camber & Caster orientation
AS5600 → Steering angle
MPX Sensor → Tyre pressure
Format data into respective structures (ADXLData / SteeringData / PressureData)
Transmit packets wirelessly via ESP-NOW to Receiver ESP32 (MAC-mapped peer connection)
Receiver ESP32 verifies packet type by data length and decodes fields
Compute vehicle alignment parameters:
Camber angle,Caster angle,Toe estimation using steering and Pressure evaluation
Compare results against predefined safe limits and assign health status for each parameter
Display live results on OLED
End (Continuous Loop)
Refresh & Update Live

4.2.1 Hardware Architecture:

The system comprises three individual sensing modules and one central receiver module: Each sensor node is powered using a 3.7 V Li-ion battery, TP4056 charging module, and MT3608 boost converter (output set to 5 V). Table 4.1 shows the hardware architecture specifications. The pressure sensor input is stabilized using a resistive voltage divider (15 kΩ + 10 Ω) to maintain safe logic levels for ESP32-C3.

Table 4.1 Hardware Architecture

Node	Sensor Type	Parame ter Measur ed	Micro contro l ler
ADXL Node	ADXL345	Camber & Caster	ESP32-C3
Steeri ng Node	AS5600	Steering Angle	ESP32-C3
Pressu re Node	MPX4115 AP	Tire Pressure	ESP32-C3
Recei ver Unit	-	Data Fusion + Display	ESP32 Dev Module

4.2.2 Firmware Development:

Node firmware was developed using the Arduino framework. Each node continuously captures real-time sensor values and encapsulates them into structured packets:

- ADXL Node → (x, y, z acceleration)
- Steering Node → (raw magnetic angle 0–4095)
- Pressure Node → (pressure_kPa, pressure_psi)

- Sensor readings are transmitted to the receiver using the ESP-NOW protocol, chosen for its low power consumption, long range, negligible latency, and ability to operate without Wi- Fi connectivity.
- A confirmation LED on every node indicates successful packet transmission.

4.2.3 Wireless Communication & Synchronization:

Since the receiver may receive packets from multiple nodes at different time intervals, the implementation supports independent data reception. Table 4.2 shows the wireless communication and synchronization results. The receiver activates computation only when packets from at least two different sensor nodes are received to avoid incorrect analysis from incomplete data.

Table 4.2 Wireless Communication & Synchronization

Parameter	Equation / Logic
Camber	Derived from tilt angle using X–Z components of ADXL
Caster	Estimated using Y–Z component relation of ADXL
Steering Angle	Raw 0–4095 mapped to 0–360°
Toe	Derived by mapping relative steering rotation to toe deviation
Tire Pressure	kPa and psi displayed directly

The receiver continuously updates three status states:

- Camber status – OK / Slight / Bad

- Caster status – OK / Check
- Toe status – OK / Toe-In / Toe-Out
- Pressure status – OK / Low / High

Final decision is computed based on sensor fusion:

- If pressure is abnormal → Pressure Issue
- Else if camber/caster/toe abnormal → Alignment Needed
- Else → Aligned

4.2.4 User Interface:

Results are displayed on a 1.3-inch OLED (I2C) in real time. The screen design incorporates Numerical values of each sensor, each parameters diagnostic state, and the last one vehicle-level decision, and recommended plan of action.

4.2.5 Physical Installation:

- In order to decouple vibrations from the wheel, the ADXL sensor node is mounted near the wheel hub using double-sided foam tape.
- The steering node is located next to the AS5600 magnet and steering column.
- While testing, the pressure node will be connected to the tire through a silicone tube; otherwise, to fix permanently it requires, a brass T-adapter.
- The receiver is also handheld and portable.

Mathematical model:

The mathematical model of the camber, caster, as well as the toe angles are to show how the wheels are positioned in relation to the vehicle body and which directly influence the stability, steering and the tire wear. Camber angle is the tilt of the wheel along X-Z plane determined using the lateral acceleration and vertical acceleration; it determines the direction of the wheel leans either inwards or outwards which affects the tire contact with the road. Caster angle can be defined as the level of the steering axis tilt in the Y-Z plane in relation with longitudinal and vertical acceleration; it defines the level of straight line tooling and the possibility to turn steering back.

The distance between the wheels and the longitudinal axis of the vehicle in the horizontal plane, which is measured by steering encoder information, is known as toe angle; it is either toe in (wheels point inwards) or toe out (wheels point outwards) which influences both handling

and tire wear. Collectively, these three parameters give a mathematical model to track and examine the wheel alignment towards safe and efficient running of the vehicles.

1. Camber Angle

Camber is the angle between the vertical axis of the wheel and the vertical axis of the vehicle when viewed from the front or rear.

Wheel Camber (tilt of wheel in X–Z plane):

$$\theta_{camber} = \arctan\left(\frac{a_x}{a_z}\right)$$

Where,

ax = acceleration in X (side tilt)

az = acceleration in Z (gravity reference)

Multiply by 180/PI to convert radians → degrees

2. Castor Angle

Caster is the angle between the steering axis (line through upper and lower ball joints or struts) and the vertical axis of the vehicle, measured from the side view.

r Wheel Caster (tilt in Y–Z plane):

$$\theta_{caster} = \arctan\left(\frac{a_y}{a_z}\right)$$

Where,

ay = acceleration in Y (longitudinal tilt)) az

= acceleration in Z (gravity reference)

Multiply by 180/PI to convert radians → degrees

3. Toe Angle

Toe is the angle between the direction the wheels point and the longitudinal axis of the vehicle, viewed from above.

$$\theta_{toe} = (angle - refSteering) \times \frac{360}{4096}$$

Where,

AS5600 gives 4096 steps/rev → each step = 0.0879°.

Compares current steering angle with calibrated reference.

Detects Toe-In (negative) or Toe-Out (positive**)**.

CHAPTER 5

CHAPTER 5

EXPERIMENTS AND RESULTS

This section shows the real-time experiment of the suggested system and the results achieved. It involves the establishment of tests, sensor measurements, and observation tables to test the alignment of the wheels and tire pressure in various conditions. The findings are compared with the accuracy, reliability and performance of the system, which is the effectiveness of the prototype to identify misalignment of tires and unhealthy tires in real time.

5.1 Experiments:

The Smart Wheel Misalignment and Tire Health Monitoring System was experimentally tested under stationary conditions of a car. Checking of the sensor nodes against proper functioning, wireless connectivity, data processing, and real-time tyre. The key objectives of the experiment were pressure and alignment parameter visualization. Three of them were wheel orientation, steering angle, pressure in tires and central handheld receiving unit.

The sensor nodes were placed at the places of measurement in order to measure the appropriate physical properties. A reliable, low latency, and sensor-node-to-sensor-node connection ESP-NOW protocol and receivers, realized wireless transmission of data. The sensor data was constantly relayed to the receiver unit where it was sorted and shown on a display. The values calculated, which include camber angle, caster angle, toe deviation and tire pressure were displayed by the receiver units. Trace repetition, wireless communication stability, and sensor measurement stability were tested in a number of ways.

5.1.1 Sensor Placement and Output Visualization

The following figures illustrate the practical implementation, sensor placement, and real- time output visualization during experimentation.

- Figure 5.1 shows the display of the OLED that shows real-time wheel alignment parameters such as camber, caster, toe and tyre pressure that the sensor nodes have sent. It confirms the successful transfer of wireless information and real-time presentation of the observed values.

Figure 5.1 OLED display showing real-time wheel alignment parameters (camber, caster, toe) and tyre pressure values

- Figure 5.2 indicates the positioning of the AS5600 magnetic rotary encoder placed next to the steering mechanism to detect the steering angle. The encoder has precise angular position information needed in estimating toe angles.

Figure 5.2 Placement of AS5600 magnetic rotary encoder for steering angle measurement

- Figure 5.3 represents the position of the ADXL345 accelerometer sensor on the wheel assembly to measure the camber and caster angles. The sensor measures the change in tilt that is computed to get the wheel alignment parameters.

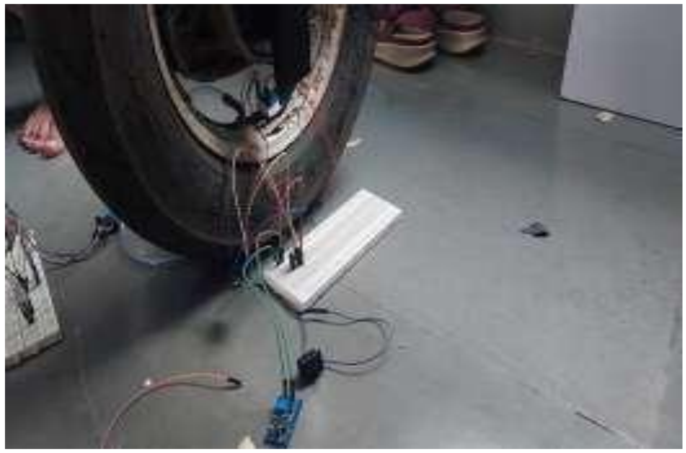


Figure 5.3 Placement of ADXL345 accelerometer sensor on wheel assembly for camber and caster measurement

- Figure 5.4 represents the position of the ADXL345 accelerometer sensor on the wheel assembly to measure the camber and caster angles. The sensor measures the change in tilt that is computed to get the wheel alignment parameters.

Figure 5.4: Placement of MPX5700AP pressure sensor connected to tyre valve using silicone tube

5.2 Results:

The data in the following trial tables is the experimental values measured on several trials under the conditions of stationary. Each trial can be identified as one cycle of measurement of wheel alignment parameters and tyre pressure.

Case 1: The representation of the experimental values in table 5.1 presents the values in the circumstances of acceptability in the alignment parameters of the wheels and the pressure of the tyre. All trials took were in normal range, and the final verdict of Aligned was retrieved and proves the appropriate work of the system.

Table 5.1 Normal Wheel Alignment with Normal Tyre Pressure

Trial No	Camber (deg)	Caster (deg)	Toe (deg)	Steering Angle(deg)	Tire Pressure(kpa)	Final Verdict
1	+0.12	+3.10	+0.08	0.5	230	Aligned
2	+0.18	+3.05	-0.05	-0.8	232	Aligned
3	-10	+3.20	+0.12	1.2	228	Aligned
4	+0.05	+2.95	-0.10	-1.0	235	Aligned
5	-0.15	+3.00	+0.06	0.0	231	Aligned
6	+0.20	+3.15	-0.08	0.7	229	Aligned
7	-0.08	+3.05	+0.10	-0.6	233	Aligned

Case 2: Table 5.2 is one of the examples in the case when the parameters of wheel alignment are at normal level and the tyre pressure is not within the given range. The system can detect this state and report a Pressure Issue even when it is put properly.

Table 5.2 Normal Wheel Alignment with Abnormal Tyre Pressure

Trial No	Camber (deg)	Caster (deg)	Toe (deg)	Steering Angle(deg)	Tire Pressure(kpa)	Final Verdict
1	+0.12	+3.1	+0.08	0.5	480	Pressure Issue
2	+0.15	+3.0	+0.05	0.0	495	Pressure Issue
3	+0.10	+2.9	+0.07	-0.4	460	Pressure Issue
4	+0.18	+3.2	+0.06	0.3	505	Pressure Issue
5	+0.11	+3.1	+0.09	-0.2	440	Pressure Issue

Case 3: Changes in camber, caster, and toe angle have been illustrated in table 5.3 and have been shifted to values that exceed the acceptable values range at normal tyre pressure. The results indicate that the system is highly effective at attending to the wheel misalignment despite the pressure of the tyres.

Table 5.3 Wheel Misalignment with Normal Tyre Pressure

Trial No	Camber (deg)	Caster (deg)	Toe (deg)	Steering Angle(deg)	Tire Pressure(kpa)	Final Verdict
1	+1.8	+4.9	+0.6	0.8	235	Misaligned
2	-1.5	+5.2	-0.7	-0.4	230	Misaligned
3	+2.2	+4.6	+0.9	0.6	240	Misaligned
4	-1.9	+5.5	-0.8	-0.2	238	Misaligned
5	+1.6	+4.8	+0.5	0.3	232	Misaligned

Case 4: As Table 5.4 shows, the steering angle values were not normal and the variation in the alignment parameters was highly significant. It is one of the conditions that the system is rightly identified as being out of alignment owing to steering anomalies.

Table 5.4 Wheel Misalignment with Steering Issue

Trial No	Camber (deg)	Caster (deg)	Toe (deg)	Steering Angle(deg)	Tire Pressure(kpa)	Final Verdict
1	+132	-148	+1.6	+68	240	Misaligned
2	-121	+155	-1.8	-72	238	Misaligned
3	+145	-162	+2.1	+74	242	Misaligned
4	-138	+149	-1.5	-65	235	Misaligned
5	+129	-141	+1.9	+70	239	Misaligned

CONCLUSION AND FUTURE SCOPE

Conclusion:

The designed system is able to exhibit the concept of designing and developing a Smart Wheel Misalignment and Tire Health Monitoring system that can be used to measure the important alignment parameters and tyre pressure when the vehicle is not in motion. The adopted system employs numerous sensor nodes to transmit the data wirelessly to a central handheld receiver governor so that the steering angle, wheel alignment and tire pressure are monitored independently. AS5600 magnetic rotary encoder is capable of sensing rotational revolution, whereas the ADXL345 acceleration sensor can be used successfully to determine the wheel inclination in order to determine the camber and caster angles. The MPX5700AP pressure sensor is capable of detecting the level of tire pressure. Each signal is analysed by the receiver unit, and mathematical calculations are performed to transform raw sensor data to the receiver into significant alignment and pressure traits.

The findings are presented immediately to an OLED display and it is easy to the user to determine the presence of tyre pressure threat or disorientation. The systems capacity to identify the normal and abnormal tire pressure limits and aligned and was experimentally confirmed to be not aligned with the wheel states. The wireless construction was found to have been stable after numerous tests and a series of measurements indicated that the sensor data were repeatable. The small size and low cost of the hardwares prove the fact that more advanced diagnostic functions can be provided and obtained without the need of complex shop equipment or the protracted installations of the lab equipment in a car. Altogether, the given project demonstrates that distributed sensing along with embedded wireless processing can provide an effective, practical, and user-friendly solution to the assessment of the vehicle condition. The system will promote early warning of deviation of alignment and tyre pressure problems, prevention maintenance, enhance awareness on road safety and conserve unnecessary tyre wear and fuel wastage. The prototype developed provides a solid basis on where further development can be made on the use of intelligent car diagnostic systems and live and dynamic car monitoring.

Future Scope:

The produced smart wheel misalignment and tyre health monitoring system is a very good starting point to the technological innovation and use in the future. The system can be extended in countless ways in case the current application is intended to provide stationary inspection and parameter identification, even to make the system more functional, usable and diagnostic. The dynamism operation system is one of the extensions and can measure the car movement. Such an increase would enable checking the state of the wheel alignment and tyre pressure of the vehicles in real-time in the real- world conditions and provide the driver with a prompt response in case the deviations have been identified.

The system is also capable of having more diagnostic parameters with the incorporation of more sensing parameters. Wheel bearer sensors, tire temperature sensors, and tire sensors will be mandatory to provide a more detailed view of the car health. Such changes would contribute to preventing maintenance abilities and competence in safety and preventive maintenance and help reveal the deterioration of components, brake problems, or hot tires in time.

The other possible enhancement would be a cloud-based interface or a mobile application. Sensor information can be sent to smartphones or cloud solutions rather than the local OLED displays, which might permit remote monitoring, history recording, predictive analytics, and alerting them of maintenance schedules. The future development can also be focused on smaller sensor nodes and enclosures that will entirely cover and protect the measurement units against dust, splashes of water, intense temperatures, and shaking. This way, the system would be better against prolonged use during autos.

One more enhancement to the system that may be made is the machine learning-based analysis that utilizes alignment and pressure data trends to predict tire deterioration and alignment drift. It would enhance the reliability of the vehicles and time of their servicing through an increase in intelligent fault prediction level of the system.

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APPENDIX

Code for Implementation

3.2.1 Steering sensor (Sender code):

```
#include <Wire.h> #include
<AS5600.h>
#include        <esp_now.h>
#include <WiFi.h>
#include  <esp_wifi.h>

// LED for status
#define LED_PIN LED_BUILTIN

// Receiver MAC Address - REPLACE WITH YOUR RECEIVER MAC!
uint8_t receiverMac[] = {0x3C, 0x8A, 0x1F, 0x7E, 0x47, 0x94};

// AS5600 sensor object
AS5600 encoder;

// Data structure struct SteeringData {
  uint16_t angle; // 0-4095 (full rotation)
} data;

bool espnowReady = false;

// LED blink patterns
```

```
void blinkLED(int times, int delayMs) { for(int i = 0; i < times; i++) {
digitalWrite(LED_PIN, HIGH); delay(delayMs); digitalWrite(LED_PIN, LOW);
delay(delayMs);
  }
}
// ESP-NOW callback
  void onSent(const uint8_t *mac, esp_now_send_status_t status) { if (status ==
ESP_NOW_SEND_SUCCESS) {     digitalWrite(LED_PIN,     HIGH);
  delay(20);
  digitalWrite(LED_PIN, LOW);
 } else {   digitalWrite(LED_PIN, HIGH);   delay(200);   digitalWrite(LED_PIN, LOW);
 }
}

void setup() {
 pinMode(LED_PIN, OUTPUT);

 // Startup - 1 long blink     digitalWrite(LED_PIN,     HIGH);     delay(1000);
digitalWrite(LED_PIN, LOW);

 // Power stabilization delay(5000);

 // Initialize I2C
 Wire.begin(4, 5); // SDA=4, SCL=5 delay(500);

 // Initialize AS5600   if (!encoder.begin()) {    //   Error:   10   fast   blinks
  while(1) { blinkLED(10, 100);     delay(2000);
   }
 }
```

```
// AS5600 OK - 2 blinks blinkLED(2, 200);

// Initialize WiFi
WiFi.mode(WIFI_OFF);
delay(500);
WiFi.mode(WIFI_STA); delay(1000);

esp_wifi_set_ps(WIFI_PS_NONE);    esp_wifi_set_max_tx_power(84);

// WiFi OK - 3 blinks blinkLED(3, 200);

// Initialize ESP-NOW if (esp_now_init() != ESP_OK) {      while(1) { blinkLED(10,
 100);                        delay(2000);
  }

esp_now_register_send_cb(onSent);

// Add peer esp_now_peer_info_t peerInfo = {}; memcpy(peerInfo.peer_addr,
receiverMac, 6); peerInfo.channel = 0;
peerInfo.encrypt = false; peerInfo.ifidx = WIFI_IF_STA;

if (esp_now_add_peer(&peerInfo) != ESP_OK) {
  while(1) {      blinkLED(10, 100);      delay(2000);
  }
}

// Ready - 4 blinks blinkLED(4, 200); espnowReady = true;
```

```
}
void loop() { if (!espnowReady) {       delay(100);     return;
  }
  // Read angle
  data.angle = encoder.rawAngle(); // Send via ESP-NOW
  esp_err_t    result    =    esp_now_send(receiverMac,    (uint8_t    *)&data,    sizeof(data));
delay(100); // Send every 100ms
}
```

3.2.2 ADXL345 sensor (Sender code):

```
#include <Wire.h>
#include <WiFi.h>
#include          <esp_now.h>
#include <esp_wifi.h>
#include <Adafruit_Sensor.h>
#include <Adafruit_ADXL345_U.h>

// LED for status indication
#define LED_PIN LED_BUILTIN // Usually GPIO8 on ESP32-C3

// Receiver MAC Address - REPLACE WITH YOUR RECEIVER MAC!
uint8_t receiverMac[] = {0x3C, 0x8A, 0x1F, 0x7E, 0x47, 0x94};

// ADXL345 sensor object
Adafruit_ADXL345_Unified accel = Adafruit_ADXL345_Unified(12345);

// Data structure to send struct ADXLData {
  float x; float y; float z;
} sensorData;
```

```
// Status tracking bool espnowReady = false; int sendCount = 0; int failCount = 0;

// LED blink patterns
void blinkLED(int times, int delayMs) { for(int i = 0; i < times; i++) {
digitalWrite(LED_PIN, HIGH); delay(delayMs); digitalWrite(LED_PIN, LOW);
delay(delayMs);
  }
}

// ESP-NOW send callback
void onSent(const uint8_t *mac_addr, esp_now_send_status_t status) { if (status ==
ESP_NOW_SEND_SUCCESS) { sendCount++; // Quick blink for success
digitalWrite(LED_PIN, HIGH); delay(20);
    digitalWrite(LED_PIN, LOW);
  } else {      failCount++; // Long blink for failure      digitalWrite(LED_PIN,
HIGH); delay(200); digitalWrite(LED_PIN, LOW);
  }
}

void setup() { // Configure LED pinMode(LED_PIN, OUTPUT);

  // Startup indication - 1 long blink digitalWrite(LED_PIN, HIGH); delay(1000);
digitalWrite(LED_PIN, LOW);

  // CRITICAL: Wait for power stabilization delay(5000);

  // Initialize I2C
  Wire.begin(4, 5); // SDA=GPIO4, SCL=GPIO5 delay(500);
```

```
// Initialize ADXL345 if (!accel.begin()) { // Error: 10 fast blinks
 while(1) { blinkLED(10, 100); delay(2000);
  }
 }

// ADXL OK - 2 blinks blinkLED(2, 200);

// Set ADXL345 range accel.setRange(ADXL345_RANGE_16_G);

// Initialize WiFi
WiFi.mode(WIFI_OFF);
delay(500);
WiFi.mode(WIFI_STA); delay(1000);

// Disable power saving for better reliability esp_wifi_set_ps(WIFI_PS_NONE);

// Set WiFi to maximum power esp_wifi_set_max_tx_power(84);

// WiFi OK - 3 blinks blinkLED(3, 200);

// Initialize ESP-NOW if (esp_now_init() != ESP_OK) { // Error: 10 fast blinks
 while(1) { blinkLED(10, 100); delay(2000);
  }
 }

// Register send callback esp_now_register_send_cb(onSent);
```

```
// Register peer (receiver) esp_now_peer_info_t peerInfo = {};
memcpy(peerInfo.peer_addr, receiverMac, 6); peerInfo.channel = 0; peerInfo.encrypt =
false; peerInfo.ifidx = WIFI_IF_STA;

  if (esp_now_add_peer(&peerInfo) != ESP_OK) {
   // Error: 10 fast blinks     while(1) {     blinkLED(10, 100);     delay(2000);
   }
  }
  // All ready - 4 blinks blinkLED(4, 200); espnowReady = true;
}

void loop() { if (!espnowReady) {     delay(100);    return;
  }

  // Read sensor data  sensors_event_t event;  accel.getEvent(&event);

  sensorData.x = event.acceleration.x; sensorData.y = event.acceleration.y; sensorData.z =
event.acceleration.z;

  // Send via ESP-NOW
  esp_err_t result = esp_now_send(receiverMac, (uint8_t*)&sensorData,
  sizeof(sensorData));

// LED indication handled by callback

delay(100); // Send every 100ms
}
```

3.2.3 MPX5700AP Pressure sensor(Sender code):

```
#include <esp_now.h>
#include <WiFi.h>
#include <esp_wifi.h>

// LED for status
#define LED_PIN LED_BUILTIN

// Pressure sensor pin (GPIO2 with voltage divider)
#define PRESSURE_PIN 2

// Receiver MAC Address - REPLACE WITH YOUR RECEIVER MAC!
uint8_t receiverMac[] = {0x3C, 0x8A, 0x1F, 0x7E, 0x47, 0x94};

// Data structure struct PressureData { float pressure_kpa; float pressure_psi;
} pdata;

bool espnowReady = false;

// LED blink patterns
void blinkLED(int times, int delayMs) { for(int i = 0; i < times; i++) {
digitalWrite(LED_PIN, HIGH); delay(delayMs); digitalWrite(LED_PIN, LOW);
delay(delayMs);
}
}

// ESP-NOW callback
```

```
void onSent(const uint8_t *mac, esp_now_send_status_t status) {     if (status ==
ESP_NOW_SEND_SUCCESS) {       digitalWrite(LED_PIN, HIGH);       delay(20);
digitalWrite(LED_PIN, LOW);
  } else {
    digitalWrite(LED_PIN, HIGH);     delay(200);    digitalWrite(LED_PIN, LOW);
  }
}

void setup() {
  pinMode(LED_PIN, OUTPUT);

  // Startup - 1 long blink         digitalWrite(LED_PIN,         HIGH);         delay(1000);
digitalWrite(LED_PIN, LOW);
  // Power stabilization delay(5000);

// Configure ADC pinMode(PRESSURE_PIN, INPUT);
analogReadResolution(12); // 12-bit ADC (0-4095) analogSetAttenuation(ADC_11db); // 0-
3.3V range

// Sensor OK - 2 blinks blinkLED(2, 200);

  // Initialize WiFi
  WiFi.mode(WIFI_OFF);
  delay(500);
  WiFi.mode(WIFI_STA); delay(1000);

  esp_wifi_set_ps(WIFI_PS_NONE);    esp_wifi_set_max_tx_power(84);
```

```
// WiFi OK - 3 blinks blinkLED(3, 200);

// Initialize ESP-NOW if (esp_now_init() != ESP_OK) {      while(1)                          {
 blinkLED(10, 100);    delay(2000);
  }
 }

esp_now_register_send_cb(onSent);

// Add peer esp_now_peer_info_t peer = {}; memcpy(peer.peer_addr, receiverMac, 6);
peer.channel = 0; peer.encrypt = false; peer.ifidx = WIFI_IF_STA;

if (esp_now_add_peer(&peer) != ESP_OK) {
  while(1) {       blinkLED(10, 100);       delay(2000);
  }
 }

// Ready - 4 blinks blinkLED(4, 200); espnowReady = true;
}

void loop() { if (!espnowReady) {          delay(100);      return;
 }

// Read ADC value (average of 10 readings for stability)     long sum = 0;    for(int i = 0; i <
10;          i++)          {          sum          +=
  analogRead(PRESSURE_PIN);
 delay(5); } int raw = sum / 10; // Convert to voltage after voltage divider float voltage_adc
= raw * (3.3 / 4095.0);
```

```
  // Reverse voltage divider (10kΩ + 15kΩ = 0.6 factor)  float voltage_sensor = voltage_adc
/ 0.6;

  // Convert to pressure using MPX5700AP formula
  // Vout = 5.0 × (0.0012858 × P + 0.04)
  // Solving for P: P = (Vout/5.0 - 0.04) / 0.0012858 float pressure_kpa = ((voltage_sensor / 5.0) -
0.04) / 0.0012858;

  // Clamp to valid range
  if (pressure_kpa < 15.0) pressure_kpa = 15.0;      if    (pressure_kpa    >    700.0)
pressure_kpa = 700.0;

  // Convert to PSI
  pdata.pressure_kpa = pressure_kpa;  pdata.pressure_psi = pressure_kpa * 0.145038;
  // Send via ESP-NOW
esp_err_t result = esp_now_send(receiverMac, (uint8_t*)&pdata, sizeof(pdata));

  delay(200); // Send every 200ms
}
```

3.2.4 ESP32(Receiver code):

```
#include <WiFi.h>
#include      <esp_now.h>
#include <esp_wifi.h>
#include <U8g2lib.h>
#include <Wire.h>
#include <math.h>

// --- FIXED LED PIN ---
#define LED_PIN 2
```

```
// OLED (I2C)
U8G2_SSD1306_128X64_NONAME_F_HW_I2C oled(U8G2_R0, /* reset=*/
U8X8_PIN_NONE);

// ------------ Structures from Senders ------------------------ struct ADXLData {
  float x;  float y;  float z;
};

struct SteeringData { uint16_t angle;
};

struct PressureData { float pressure_kpa; float pressure_psi;
};

// Received data holders
ADXLData adxl;
SteeringData steering; PressureData pressure; bool gotADXL = false; bool gotSteering =
false; bool gotPressure = false;

// ----- Calibration + Geometry params ----- float steer_calib = 0.0f; // set at runtime if
needed float wheel_roll_baseline = 0.0f; // set at runtime if needed
float steerToToeFactor = 0.02f; // empirical mapping: toe(deg) = steerRel(deg) * factor

// thresholds (tunable)
const float CAMBER_OK = 0.5f;  // deg const float CASTER_MIN = 2.0f; // deg const
float CASTER_MAX = 5.0f; // deg const float TOE_OK = 0.5f; // deg const float
PRESSURE_LOW_KPA = 180.0f; const float PRESSURE_HIGH_KPA = 260.0f;
```

```
// vehicle geometry (for ackermann calc) - tune per vehicle float L = 2.44f; // wheelbase (m) float t
= 1.48f; // track width (m)

// util
float rad2deg(float r){ return r * 57.29577951308232f; } float deg2rad(float d){ return d *
0.017453292519943295f; } float normalize180(float a){ while(a > 180.0f) a -= 360.0f;
while(a <= -180.0f) a += 360.0f; return a; }

// ackermann inner angle expected (deg)
float computeAckermannInnerDeg(float steer_deg_effective) { float srad =
deg2rad(steer_deg_effective); if (fabsf(tanf(srad)) < 1e-6f) return 0.0f; float R = L /
tanf(srad); float inner = atan2f(L, (R - (t/2.0f))); return rad2deg(inner); }

// compute camber from ADXL (deg) float computeCamber(const ADXLData &a) {
  return rad2deg(atan2f(a.x, a.z));
}

// compute caster approximation from ADXL (deg) float computeCaster(const ADXLData &a)
{ return rad2deg(atan2f(a.y, a.z));
}

// AS5600 raw 0..4095 -> degrees 0..360 float steeringRawToDeg(uint16_t raw) {      return
((float)raw * 360.0f / 4096.0f);
}

// friendly status strings
String statusCamber(float camber) { if (fabsf(camber) <= CAMBER_OK) return "OK"; if
(fabsf(camber) <= 1.0f) return "Slight"; return "Bad";
```

```
}
String statusCaster(float caster) { if (caster >= CASTER_MIN && caster <=
 CASTER_MAX) return "OK"; return "Check";
}
String statusToe(float toe) {   if (fabsf(toe) <= TOE_OK) return "OK";   return (toe > 0) ?
"Out" : "In";
}
String statusPressure(float p) {
 if (p < PRESSURE_LOW_KPA) return "Low"; if (p > PRESSURE_HIGH_KPA) return
"High"; return "OK";
}

// ________________________________________.
// NEW CALLBACK FORMAT FOR ESP32 v3.0.7
// ________________________________________.
void onReceive(const esp_now_recv_info_t *info, const uint8_t *incomingData, int len) {
 // Identify data type by length if (len == sizeof(ADXLData)) { memcpy(&adxl,
incomingData, sizeof(adxl)); gotADXL = true;
 }
 else if (len == sizeof(SteeringData)) {
  memcpy(&steering, incomingData, sizeof(steering));     gotSteering = true;
 }
 else if (len == sizeof(PressureData)) {
  memcpy(&pressure, incomingData, sizeof(pressure));     gotPressure = true;
 }

 // Blink on receive
 digitalWrite(LED_PIN, HIGH); delay(20); digitalWrite(LED_PIN, LOW);
}
```

```
void drawHeader() {      oled.setFont(u8g2_font_ncenB08_tr);      oled.drawStr(0, 10,
"WheelGuard"); oled.setFont(u8g2_font_5x8_tr); oled.drawStr(86, 8, "v1");
 // separator oled.drawHLine(0, 12, 128);
}

void showSummary(const String &verdict, const String &advice) {
 // bottom two lines: verdict + short advice  oled.setFont(u8g2_font_7x13B_tr);
 // try to center verdict  int x = 0;  // truncate if too long
 char bufV[32]; bufV[0]=0; strncpy(bufV, verdict.c_str(), 31);    oled.drawStr(0, 58, bufV);
oled.setFont(u8g2_font_5x8_tr);
 char bufA[40]; bufA[0]=0; strncpy(bufA, advice.c_str(), 39);     oled.drawStr(0, 64-6,
 bufA);
}

void setup() { Serial.begin(115200); delay(1000);

 // LED
 pinMode(LED_PIN, OUTPUT);

 // OLED       oled.begin();   oled.clearBuffer();     oled.setFont(u8g2_font_6x12_tr);
oled.drawStr(0, 15, "Receiver Booting..."); oled.sendBuffer();

 // WiFi INIT
 WiFi.mode(WIFI_STA); esp_wifi_set_ps(WIFI_PS_NONE);

 // ESP-NOW INIT if (esp_now_init() != ESP_OK) { Serial.println("ESP-NOW init
failed!"); while (1);
 }
```

```
  // Register callback  esp_now_register_recv_cb(onReceive);

  oled.clearBuffer(); oled.drawStr(0, 15, "Receiver Ready!");  oled.sendBuffer();
}

void loop() {

  // compute metrics only when we have at least one packet
  float camber = 0.0f, caster = 0.0f, toe_est = 0.0f, steer_rel = 0.0f, ack_expected = 0.0f;
  float
pressure_kpa = 0.0f;
  String camberS = "N/A", casterS = "N/A", toeS = "N/A", pressS = "N/A";

  if (gotADXL) {
     camber = computeCamber(adxl) - wheel_roll_baseline; // relative    caster =
computeCaster(adxl);   camberS = statusCamber(camber);   casterS = statusCaster(caster);
  }

  if (gotSteering) {
   float steer_deg = steeringRawToDeg(steering.angle);     steer_rel = normalize180(steer_deg
- steer_calib);   toe_est   =   steer_rel   *   steerToToeFactor;   ack_expected   =
   computeAckermannInnerDeg(steer_rel); toeS = statusToe(toe_est);
  }

  if (gotPressure) {
   pressure_kpa = pressure.pressure_kpa;     pressS     =
  statusPressure(pressure_kpa); }
```

```
// final verdict & advice
String finalVerdict = "ALIGNED"; String advice = "No action"; if (pressS != "OK") {
  finalVerdict = "PRESSURE ISSUE";
 advice = (pressure_kpa < PRESSURE_LOW_KPA) ? "Inflate tyre" : "Check tyre";
    } else if (camberS != "OK" || casterS != "OK" || toeS != "OK") {          finalVerdict =
"ALIGNMENT        NEEDED";
  advice = "";
  if (camberS != "OK") advice += "Adjust camber; ";          if (casterS != "OK") advice
+= "Check caster; ";      if (toeS != "OK") advice += "Adjust toe; ";
 }
// ----- OLED layout (clean version) ----- // ----- OLED layout (clean + small font) -----
oled.clearBuffer(); drawHeader(); // same header

 oled.setFont(u8g2_font_5x8_tr); // SMALL FONT (no bold) char

 buf[40];

 // --- Camber ---
   snprintf(buf, sizeof(buf), "Camber: %+0.2f", camber);          oled.drawStr(0, 20, buf);
oled.drawStr(95, 20, camberS.c_str());

 // --- Caster ---
 snprintf(buf,    sizeof(buf),    "Caster:         %0.2f", caster);        oled.drawStr(0,
  30, buf); oled.drawStr(95, 30, casterS.c_str());

 // --- Steering angle --- if (gotSteering) { snprintf(buf, sizeof(buf), "Steer: %0.1f",
normalize180(steeringRawToDeg(steering.angle))); oled.drawStr(0, 40, buf);
```

```
snprintf(buf, sizeof(buf), "Toe: %0.2f", toe_est);          oled.drawStr(0, 50, buf);
} else { oled.drawStr(0, 40, "Steer: --"); oled.drawStr(0, 50,
  "Toe: --");
}

// --- Pressure --- if (gotPressure) {    char    pbuf[20];    snprintf(pbuf,    sizeof(pbuf),
  "P:%0.0f kPa", pressure_kpa);          oled.drawStr(88, 20, pbuf);
oled.drawStr(88, 30, pressS.c_str());
} else { oled.drawStr(88,
  20, "P: --");
}
// --- Ackermann --- char ackbuf[30];
snprintf(ackbuf, sizeof(ackbuf), "AckExp:%0.2f", ack_expected); oled.drawStr(64, 40,
ackbuf); // --- Final verdict --- oled.drawHLine(0, 54, 128); oled.drawStr(0, 62,
finalVerdict.c_str()); oled.sendBuffer();
}
```

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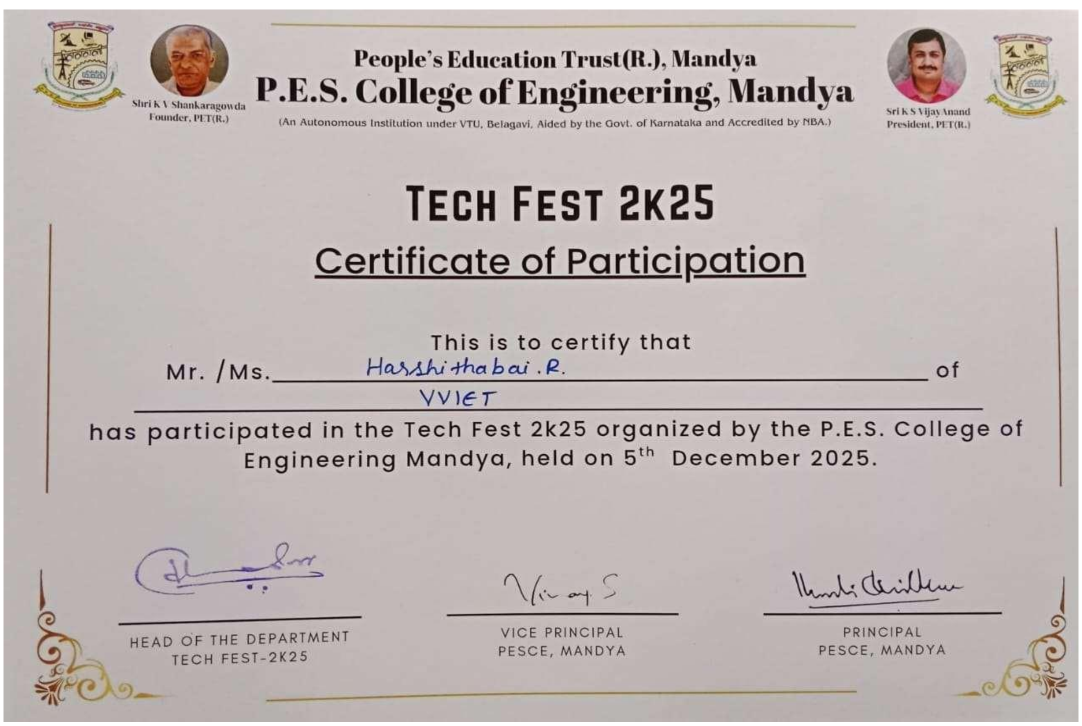

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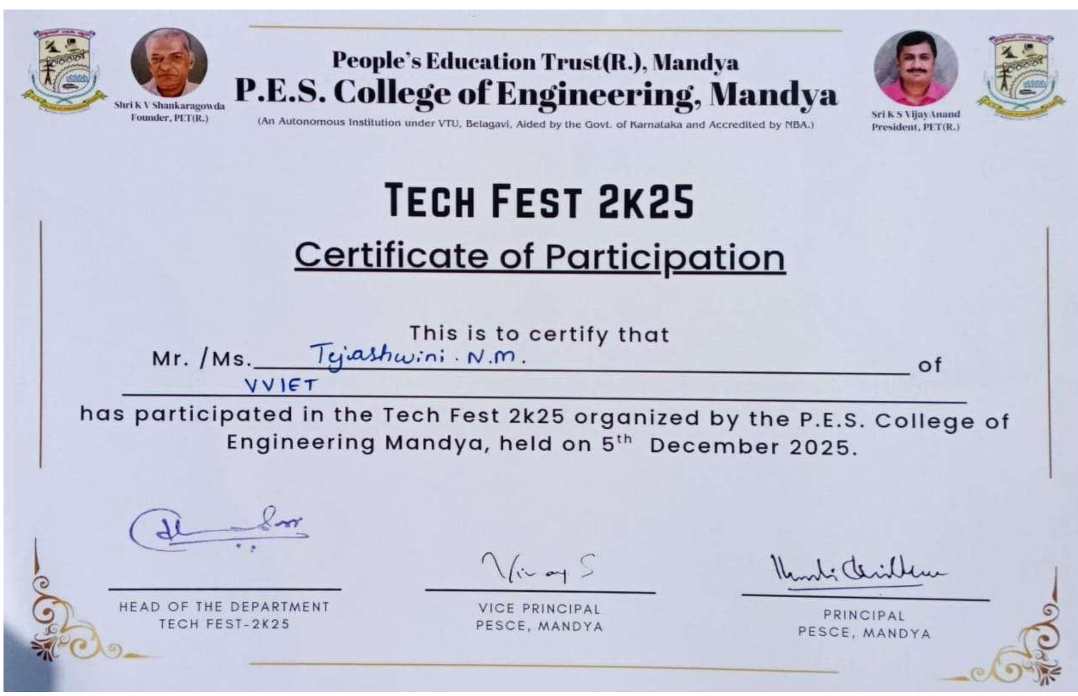

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